

Justification for Other than Full and Open Competition

“Source Selection Information – see FAR 2.101 and 3.104”

1. Identification of the agency and contracting activity.

- a. **Federal agency and contracting activity.** U.S. Department of Health and Human Services (HHS), National Institutes of Health (NIH)
- b. **Sponsoring organization.** National Institute of Allergy and Infectious Diseases (NIAID), Division of Intramural Research (DIR), Center for Human Immunology (CHI)
- c. **Project Officer information.**
 - **Project Officer name.** [REDACTED]
 - **Mailing address.** [REDACTED]
 - **E-mail address.** [REDACTED]
 - **Telephone number.** [REDACTED]

2. Nature and/or description of the action being approved.

a. Acquisition purpose and objectives.

The purpose of this acquisition is to purchase 59 SomaScan 7K and 11K assay kits for the Center for Human Immunology (CHI) for the ability to analyze clinical samples (serum, plasma, cell lysates, tissue homogenates, urine, and cerebrospinal fluid) with highly multiplexed analytes and high throughput proteins. The kits will be used to measure 11,000 proteins across 10 logs of concentration dynamic range in less than 150 uL of human biological samples with median %CV<5% across all analytes. Currently, there are numerous CHI projects and NIH fee-for-service customers from various institutes and centers in the NIH Intramural Research Program (IRP) that are awaiting their samples to be run on the platform for their clinical protocols.

b. Project background.

The CHI develops and applies systems biology approaches combining computation, modeling, and experimental data to study the immune system at the organismal, cellular, and molecular levels. The proteomic data generated from the samples that will be run on the assay from this acquisition will be coupled to information from orthogonal experimental techniques to uncover a systems immunology-based model of disease. The government requires SomaScan 7K and 11K assay kits, manufactured by SomaLogic Operating Co. (SomaLogic), to measure 11,000 proteins across 10 logs of concentration dynamic range in <150 uL of human biological samples (Serum, Plasma, CSF, Urine, Cell Lysate, Tissue Homogenate, etc.) with median %CV <5% across all

analytes. The data generated from the SomaScan 7K and 11K assay kits must be directly comparable to previously generated data that utilized the SomaScan 7K and 11K assay kits as many of the studies utilizing this requirement are continuations of longitudinal studies, and any differences in data comparability may result in detrimental impacts such as the creation of data artifacts that cannot be overcome as well as potential damage to study data integrity where remediation would not be possible.

3. Description of the supplies or services required to meet the agency's needs (including the estimated value).

a. Project title. Acquisition of SomaScan 7K and 11K Assay Kits

b. Project description.

Currently, the CHI performs systems biology research on several biological matrices using the SomaScan Assay system. The SomaScan Assay is performed using proprietary SomaScan assay kits on dedicated instrumentation that was previously purchased (SomaScan v4.1 Assay runs on a Tecan Fluent Liquid Handling platform) and is integrated into CHI's laboratory infrastructure. It enables highly multiplexed (>11,000 analytes) and high throughput (96 samples per kit) analysis of 11,000 proteins in clinical samples (e.g. human serum, plasma, cell lysates, tissue homogenates, urine, and cerebrospinal fluid). These assay kits are needed for projects involving work for ongoing clinical trials and longitudinal studies where the efficacy of the therapy to the patients would be at risk without the kits. With the SomaScan assay kits, it would provide investigators across the NIH intramural research community access to a unique set of capture reagents called "Slow Off rate Modified Aptamers (SOMAmers)" to measure the relative abundance of proteins in a multitude of biological matrices including human serum, plasma, cerebrospinal fluid, urine as well as many others. The SomaScan kits are used to search for biomarker signatures in a variety of biomedical applications, such as those predicting patient outcomes or characterizing responses to interventions and therapies at the NIH. Moreover, the CHI has successfully harnessed this technology since 2014 for use on CHI projects, as well as offering it to the greater NIH intramural community through its Technology Partnership Program. The CHI is the only laboratory and center at the NIH that is authorized to purchase SomaScan assay kits as they have two trained and certified laboratory operators to run the SomaScan Assay so that investigators at other institutes and centers at the NIH can access this technology. Therefore, the acquisition of the SomaScan assay kits is critical as it will provide significant advancement in the information that can be extracted from these samples.

• **Requirement type.**

- ☐ Research & development (R&D)
- ☐ R & D support services
- ☐ Support services (non-R&D)
- ☒ Supplies/equipment
- ☐ Information technology (IT)
- ☐ Construction

- ☐ Architect-engineer (A & E) services
- ☐ Design-build
- ☐ Other (specify): _____

• **Type of action.**

- ☒ New requirement
- ☐ Follow-on
- ☐ Other (specify): _____

• **Proposed contract/order type.**

- ☒ Firm-fixed-price
- ☐ Other fixed-price (specify, e.g., fixed-price award-fee, fixed-price incentive-fee): _____
- ☐ Cost-plus-fixed-fee
- ☐ Other cost reimbursement (specify, e.g., cost-plus-award-fee, cost-plus-incentive-fee): _____
- ☐ Time and materials
- ☐ Indefinite delivery (specify whether indefinite quantity, definite quantity, or requirements): _____
- ☐ Other (specify): _____
- ☐ Completion Form ☐ Term form

• **Acquisition identification number.** AMBIS #2244738

c. Total estimated dollar value and performance/delivery period.

The performance period is estimated to be from 3/31/2025 to 3/30/2026 and the total estimated dollar value is XXXXXXXXXX

4. Identification of the statutory authority permitting other than full and open competition.

☒ This acquisition is conducted under the authority of 41 United States Code (U.S.C.) 3304(a)(1) as set forth in Federal Acquisition Regulation (FAR) 13.106-1(b)(2) and further noted in FAR 13.501(a)(1)(ii).

☐ This acquisition is conducted under the authority of section 4202 of the Clinger-Cohen Act of 1996.

☐ This acquisition is conducted under the authority of the Services Acquisition Reform Act of 2003 (41 U.S.C. 428a).

5. Demonstration that the proposed contractor(s) unique qualifications or the nature of the acquisition requires use of the authority cited.

a. Name and address of the proposed contractor(s).

SomaLogic Operating Co. (SomaLogic)
2945 Wilderness Place

Boulder, CO 80301

b. Nature of the acquisition and proposed unique qualifications of the contractor(s).

The purpose of this acquisition is to procure 59 SomaScan7K and 11K assay kits to run samples to generate proteomic data for various studies and projects for NIAID and other institutes and centers at the NIH on the existing SomaScan V4.1 Assay, which runs on a Tecan fluent liquid handling platform. These assay kits have been integrated into longitudinal studies and are required for the final step in those projects that have spanned across multiple years dating back to 2015. Moreover, they are critical for work on ongoing clinical trials as well. The assay kits are unique in that they can rapidly and simultaneously measure more than 11,000 low and high-abundance proteins across 10 logs of dynamic range (concentration) with very high accuracy and reproducibility (median %CV of replicate measurements less than 5%) and low sample volume input requirements (less than 55uL). They enable highly multiplexed (greater than 11,000 analytes) and high throughput (96 samples per kit) analysis of 11,000 proteins in clinical samples (e.g., human serum, plasma, cell lysates, tissue homogenates, urine, cerebrospinal fluid, etc.).

Other technologies in this field, including other affinity-based protein assays, targeted protein assays, or mass-spectrometry based proteomic detection assays and technologies can only do some, but not all, of the capabilities of the SomaScan Assay. Furthermore, alternative technologies measure fewer proteins than the SomaScan Assay, do not have the ability to detect low abundance proteins (e.g. cytokines) as is possible with the SomaScan Assay, require patient sample volumes that far exceed what is required with the SomaScan Assay and more sample volume than is available to be analyzed, have poorer reproducibility than the SomaScan Assay leading to non-detection of biological signal changes, have sample throughputs far lower than that of the SomaScan Assay leading to increased wait times, labor costs, and delayed results; and the cost to run samples is far greater with other technologies than to run them on the SomaScan Assay. In addition to being independently verified in peer-reviewed scientific papers, the capabilities of the SomaScan Assay have also been verified by direct testing in our lab, and is therefore, the reason why the SomaScan Assay was selected in our laboratory for this type of work.

The proteomic data generated from the samples run on the assay from this acquisition will be coupled to information from orthogonal experimental techniques to uncover a systems immunology-based model of disease. The data generated from the assay kits must be directly comparable to previously generated data from utilization of the SomaScan 7K and 11K assay kits. Many studies utilizing this requirement are continuations of longitudinal studies, and any differences in data comparability could create data artifacts that cannot be overcome as well as potentially damage study data integrity. If alternative technologies and kits, other than the existing SomaScan Assay and the proprietary SomaScan 7K and 11K assay kits, are used for this work, the samples previously analyzed with the SomaScan Assay may not be

directly comparable to those analyzed on other platforms. This could necessitate significant costs and resources to re-analyze these samples. Furthermore, some of the previously analyzed samples have no remaining sample volume. Additionally, alternative platforms may produce inferior, unreliable, and incomplete sample measurements, as well as the cost per sample analysis would be higher for the Government should alternative platforms be utilized.

SomaLogic is the only manufacturer that produces compatible kits for the SomaScan Assay technology that meets the needs of CHI. The SomaScan Assay Kits are designed to integrate exclusively with the existing equipment and software which are available only from SomaLogic. Adopting other technologies would entail substantial capital expenditure, potentially exceeding \$1 million, and a significant amount of time to acquire the necessary dedicated instrumentation as CHI's laboratory infrastructure is currently designed to process the SomaScan Assay specifically and the proprietary SomaScan assay kits. CHI has already invested in excess of \$1.5 million on equipment to run the SomaScan Assay and certified two (2) laboratory operators to run the SomaScan Assay and kits, as required by SomaLogic to procure the kits. The SomaScan assay kits cannot be used independently from the SomaScan Assay system, nor can alternative kits be run on the SomaScan Assay system. If alternative platforms or kits were utilized, it could necessitate the Government to invest further financial resources and time that would delay the progress of the existing projects. Therefore, it is essential to procure the SomaScan assay kits from SomaLogic.

- 6. A description of efforts made to ensure that offers are solicited from as many potential sources as is practicable, including whether a notice was or will be publicized as required by subpart 5.2 and, if not, which exception under 5.202 applies.**

As required by FAR Subpart 5.2, a Notice of Intent (NOI) to Sole Source (ID: NOI-NIAID-25-2244738) was posted on the Federal Contract Opportunity site (www.SAM.gov). The notice announced the Government's intent to procure fifty-nine (59) SomaScan 7K and 11K assay kits on a non-competitive basis. Responsible sources who could provide comparable products were provided an opportunity to respond with a capability statement. If a reasonable capability statement from another source was provided, the intent was then to post a Request for Quotations (RFQ) on a full and open competition basis. NIAID received two (2) responses from SomaLogic and LC Industries (LCI).

- 7. Determination by the Contracting Officer that the anticipated cost/price to the Government will be fair and reasonable.**

The techniques identified at FAR 13.106-3(a)(2) will be used to ensure aware will be at a fair and reasonable price. NIAID will consider the Independent Government Estimate, market research, comparison with similar items in the marketplace, and quoted prices found reasonable on previous purchases as determining factors for price reasonableness. A final determination by the Contracting Officer will be conducted prior to award of a purchase order.

- 8. Description of the market research conducted (see FAR Part 10) and the results, or a statement of the reasons market research was not conducted.**

The Program Office conducted the initial market research by surveying the field of proteomics to determine if there are other prospective sources of the SomaScan assay kits. This market research was performed through multiple methods. These methods include reviewing scientific literature including reviewing technical articles in major scientific publications. In addition, staff have been able to attend scientific conferences in the field of proteomics and had the ability to view presentations related to scientific results using SomaScan Kits. The last form of market research that was performed is related to data that has been generated at the NIH related to using SomaScan assay kits to enable highly multiplexed (>11,000 analytes) and high throughput (96 samples per kit) analysis of 11,000 proteins in clinical samples (serum, plasma, cell lysates, tissue homogenates, urine, and cerebrospinal fluid). Based on the Program Office's market research, it appears that SomaLogic is the only source that manufactures and distributes compatible kits that meets the requirement.

Additional market research was conducted by the Contract Specialist via internet searches as well as searches through the Federal Procurement Data System (FPDS) for previous awards, GSA Advantage, AMBIS, and SAM.gov. NIAID issued a Notice of Intent (NOI) to Sole Source (ID: NOI-NIAID-25-2244738) on SAM.gov on January 7, 2025 to announce the Government's intent to procure the SomaScan 7K and 11K assay kits on a non-competitive basis and to reconfirm there are no other sources for the required assay kits. Responsible sources who could provide comparable products were provided an opportunity to respond with a capability statement by January 22, 2025. NIAID received two (2) responses from SomaLogic and LC Industries (LCI). SomaLogic provided an updated quote while LCI provided their capability statement. In review of the

responses, the Program Office confirmed that SomaLogic appears to be capable of satisfying the requirement while LCI's response did not offer a product but instead, offered a service to package components into kits. Such service did not appear to be within scope of CHI's requirement and is not a valid alternative to the SomaScan assay kits and therefore, did not appear to be capable of satisfying the Government's needs.

9. Any other facts supporting the use of other than full and open competition.

CHI is the only laboratory or center at the NIH that is authorized to purchase SomaScan Assay Kits so that investigators across the intramural research community can access this technology at NIH. The CHI has invested, at great cost to the Government, in training and certifying two (2) laboratory operators to run SomaScan Assay kits (required by SomaLogic for authorization). Since 2014, data for approximately 10,000 samples has been generated using SomaScan assay kits at the CHI. CHI bioinformaticians have also created numerous webtools, databases, and analysis pipelines specific to the unique data generated at CHI using SomaScan Assay kits. Additionally, the CHI have invested greater than \$1.5 million in specialized dedicated equipment and the training and certification of two (2) laboratory operators to operate the SomaScan Assay system with the proprietary SomaScan Assay Kits at CHI.

10. Listing of sources, if any, that expressed, in writing, an interest in the acquisition.

NIAID received two (2) responses from SomaLogic and LCI in response to the NOI posting (ID: NOI-NIAID-25-2244738).

11. Statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the required supplies or services.

It is the policy of the National Institute of Allergy and Infectious Diseases to promote the use of competitive procedures whenever and wherever possible. In an effort to remove or overcome barriers to competition in the future, the market will be scanned frequently for new companies and products that could satisfy the Government's requirement. The agency will continually re-evaluate the existing equipment's functionality and will continue to research the various technologies, manufacturers, and products that could be in the Government's best interest. The current requirement calls for the procurement of fifty-nine (59) SomaScan 7K and 11K assay kits from SomaLogic.

12. Program office certification.

This is to certify that the portions of this justification that have been developed by the undersigned program office personnel, including supporting information and/or data verifying the Government's minimum needs, schedule requirements and other rationale for other than full and open competition, are accurate and complete.

Official	Name & Title	Signature	Date
Project Officer	[REDACTED]	[REDACTED]	[REDACTED]
Project Officer's Immediate Supervisor	[REDACTED]	[REDACTED]	[REDACTED]
Head of the Sponsoring Program Office	[REDACTED]	[REDACTED]	[REDACTED]

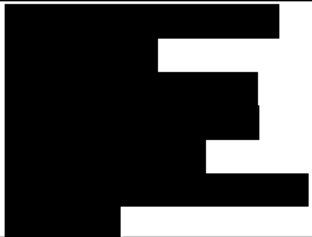
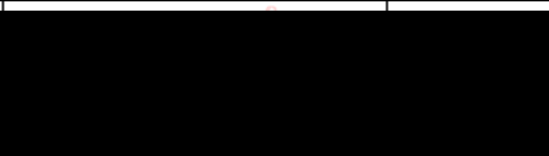
13. Contracting Officer certification.

This is to certify that the justification for the proposed acquisition has been reviewed and that to the best of my knowledge and belief the information and/or data provided to support the rationale and recommendation for approval is accurate and complete.

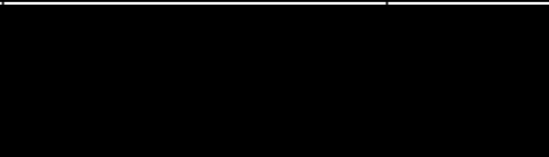
Official	Name & Title	Signature	Date
Contracting Officer	[REDACTED]	[REDACTED]	[REDACTED]

14. Chief of the Contracting Office and Head of the Contracting Activity signature(s).

Official	Name & Title	Signature	Date
Chief of the Contracting Office	[REDACTED]	[REDACTED]	[REDACTED]
	[REDACTED]	[REDACTED]	[REDACTED]

Head of the Contracting Activity			

15. Competition Advocate signature.

Official	Name & Title	Signature	Date
Competition Advocate			

16. HHS Senior Procurement Executive signature.

Official	Name & Title	Signature	Date
HHS Senior Procurement Executive	N/A		